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6 Ultraviolet radiation of wavelength 122 nm is used to illuminate the cathode in the vacuum tube as shown in Fig. 6.1.

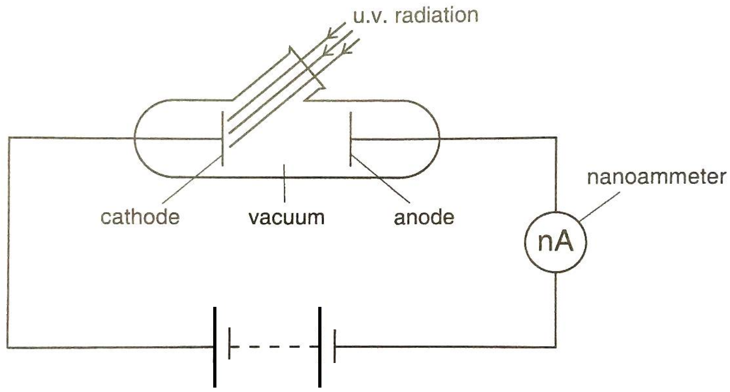

Fig. 6.1

(a) Photoelectrons are emitted from the cathode and collected at the anode. With the anode made negative and the cathode positive, some photoelectrons can still reach the anode, and by varying the battery's e.m.f, a graph of current against e.m.f. can be plotted as shown in Fig. 6.2.

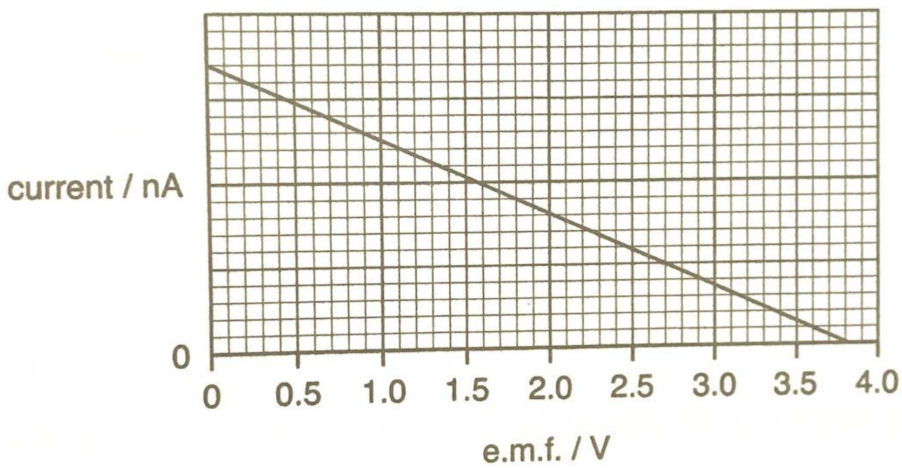

Fig. 6.2

(i) Explain why some photoelectrons are still able to reach the negative anode.

Photoelectrons emitted have kinetic energies ranging from zero to a maximum values. Those photoelectrons that **have enough energy to do work against the electric field** will be able to reach the anode.

....

.... [1]

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(ii) Calculate the maximum speed the photoelectrons can have when they leave the cathode.

Loss in KE = Gain in EPE
$\frac{1}{2} mv_{max}^2 = eV_s$
$\frac{1}{2} (9.11 \times 10^{-31}) V_{max}^2 = (1.6 \times 10^{-19})(3.8)$ [1]
$V_{max} = 1.16 \times 10^6$ m s^{-1} [1]

maximum speed = …………………………………… m s^{-1} [2]

(iii) Calculate the work function of the metal used in the cathode.

$hf = \phi + eV_s$
$\phi = (6.63 \times 10^{-34})(3 \times 10^8)/(122 \times 10^{-9}) - 1.6 \times 10^{-19}(3.8)$ [1]
$= 1.02 \times 10^{-18}$ J [1]

work function = ………………………………… J [2]

(b) The photocurrent I for different potential difference V between the cathode and the anode was measured. The experiment was then repeated using ultraviolet radiation of the same wavelength but of different intensity.

The series of graphs of I against V are shown in Fig. 6.3.

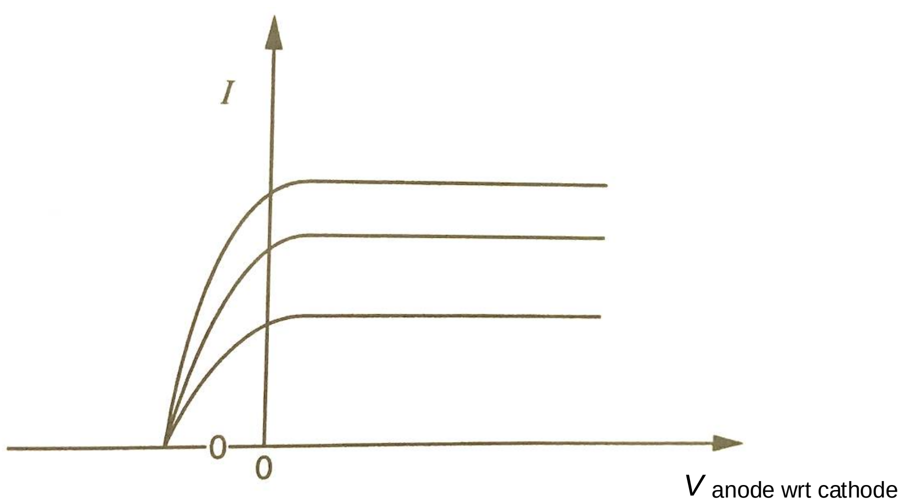

Fig. 6.3

(i) State and explain which feature of this graph could not be explained using the wave theory of light.

The stopping potential is independent of the intensity of the radiation [1]. According to wave theory, the higher the intensity, the more energy the light would possess which would cause the **photoelectrons emitted to have higher kinetic energy**, and hence **a higher stopping potential to bring them to rest** [1].

[2]

(ii) Explain why the maximum kinetic energy of photoelectrons is independent of intensity whereas the photoelectric current is proportional to intensity of the light.

When electromagnetic radiation is irradiated on the metal surface, the energy is delivered in packets known as photons. The energy of each photon energy is *hf* (where h is Planck's constant and f is the frequency of radiation.) [B1]

Photons interact with the electrons on a 1-to-1 basis such that the maximum kinetic energy of the emitted photoelectron, $KE_{max} = hf - \phi$ is dependent only on the frequency of the radiation and ϕ, the work function of the metal. [B1]

For light of a constant frequency, intensity is directly proportional to the rate of incidence of photons which is in turn directly proportional to the rate of photoelectrons emitted, and hence the photoelectric current. [B1]

[3]

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7 An induced nuclear fission reaction maybe represented by the equation